

ANALYSTLAB AFRICA

Experience Lab Internship Programme



HealthConnect Clinic Risk and Dependency Register

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Project: HealthConnect Clinic Experience Lab – Improving Patient Appointment Attendance and
Healthcare Support Using Data and AI
Track: Data Science

Prepared for the AnalystLab Africa Experience Lab
Week 4 Submission – Supporting Document

Table of Contents

1. Introduction
 2. Purpose and Scope
 3. Risk Register
 4. Dependency Register
 5. Review and Escalation Approach
- References

1. Introduction

This Risk and Dependency Register is a supporting document for the Data Science track's Week 4 work on the HealthConnect Experience Lab. It consolidates the risks and cross-track dependencies identified while defining the no-show prediction machine learning problem, and complements the Machine Learning Problem Definition Document and Week 4 Project Summary already produced for this track.

Maintaining a live risk and dependency register is standard project management practice and is recommended by the Project Management Institute's guidance on identifying, analysing and responding to project risk throughout a project's life cycle (Project Management Institute [PMI], 2021). This register should be reviewed and updated at the start of each subsequent week of the Experience Lab.

2. Purpose and Scope

The purpose of this register is to record known risks to the Data Science track's proposed approach, and dependencies between this track and the other HealthConnect Experience Lab tracks (Project Management, Data Analytics, Machine Learning Engineering, and Generative AI), so that they can be actively monitored rather than left implicit.

This document covers Week 4 items only. It is expected to be extended, rather than replaced, in subsequent weeks as the project moves through the Analysis and Solution Design, Development, Testing and Refinement, and Final Presentation stages described in the Week 4 brief.

3. Risk Register

ID	Risk Description	Category	Likelihood	Impact	Mitigation	Owner	Status
R1	waiting_time _minutes has an ambiguous definition and may represent post-visit information, introducing data leakage if used as a model input.	Data quality / leakage	Medium	High	Clarify with data owner before Week 5; provisionally excluded from the feature set.	Data Science	Open
R2	Excluding Cancelled appointments from the target definition may not match the clinic's actual operational priorities.	Scope / requirements	Low	Medium	Confirm target-variable definition with stakeholders early in Week 5.	Data Science / Project Management	Open
R3	A trained model could perform unevenly across demographic subgroups (e.g., gender, age group), leading to	Ethical / responsible AI	Medium	High	Conduct a subgroup performance review before any deployment is considered, consistent with the NIST AI	Data Science	Open

	inequitable outreach.				Risk Management Framework (Tabassi, 2023).		
R4	The dataset is fictional and synthetic and may not fully reflect real clinical scheduling behaviour.	Data / external validity	High	Medium	Treat findings as illustrative of methodology; state this limitation explicitly in all reporting.	Data Science	Accepted
R5	Without version control and a reproducible pipeline, the modelling work could accumulate technical debt that is costly to maintain.	Technical / engineering	Medium	Medium	Adopt Git-based version control and a documented, reproducible pipeline; coordinate with the Machine Learning Engineering track (Sculley et al., 2015).	Data Science / ML Engineering	Open
R6	Misalignment between this track's timeline and other tracks' deliverables could delay the shared HealthConnect project.	Project / schedule	Medium	Medium	Weekly cross-track sync and shared documentation, coordinated through the Project Management track.	Project Management	Open

4. Dependency Register

ID	Dependency Description	Type	Related Track(s)	Status
D1	The Machine Learning Engineering track's system architecture depends on the target variable and candidate feature set defined in this track's Week 4 output.	Downstream (outbound)	Machine Learning Engineering	Delivered (Week 4 documentation available)
D2	This track's modelling approach depends on clarification of the waiting_time_minutes field from the data owner or mentor.	Blocking (inbound)	Data provider / mentor	Open
D3	The interpretation of 'Cancelled' as operationally distinct from 'No-Show' depends on the HealthConnect Clinic Knowledge Base, a	Shared resource	Generative AI	Satisfied for Week 4

	resource primarily owned by the Generative AI track.			
D4	This track's Week 5–8 timeline depends on the Work Breakdown Structure and High-Level Timeline produced by the Project Management track.	Cross-track (inbound)	Project Management	Pending (see accompanying Project Plan)
D5	A future subgroup fairness review depends on the continued availability of demographic fields (gender, age_group) in the final modelling dataset.	Data dependency	Data Science (internal)	Satisfied (fields present in current dataset)

5. Review and Escalation Approach

This register will be reviewed at the start of each week of the Experience Lab. New risks or dependencies will be added as they are identified, and existing entries will be updated to reflect their current status (Open, In Progress, Accepted, Closed).

- **Risks rated High impact (R1, R3):** escalated immediately to the assigned mentor and, where relevant, to the Project Management track, rather than waiting for the next scheduled review.
- **Risks rated Medium or Low impact (R2, R4, R5, R6):** monitored and reviewed weekly, with escalation only if likelihood or impact increases.
- **Blocking dependencies (D2, D4):** tracked closely, since they have the potential to delay the Week 5 modelling work if not resolved.

References

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- Sculley, D., Holt, G., Golovin, D., Davydov, E., Phillips, T., Ebner, D., Chaudhary, V., Young, M., Crespo, J.-F., & Dennison, D. (2015). *Hidden technical debt in machine learning systems*. In Advances in Neural Information Processing Systems 28 (NeurIPS 2015) (pp. 2503–2511). <https://papers.nips.cc/paper/5656-hidden-technical-debt-in-machine-learning-systems>
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